

Tuaan 6 – Lap trinh full stack

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1.http protocol

Http is the commutation protocol between Client and Server
(HyperText Transfer Protocol)

in full stack applications:

React Frontend

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http request

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Spring Boot backend

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http response

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React Frontend

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http request

GET /products

hosst: localhost:8080

Content-Type: application/json

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Important Parts:

Method Action type

URL API address

Header Metadata

Body Send data

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http methods:

GET (read data); POST(create data); PUT(update data); DELETE(remove data)

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Status Code:

200 (Success); 201 (Created); 400 (Bad Request); 401(Unauthorized)

404 (Not found); 500 (Server Error)

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Stateless

http is stateless

The server does not remember previous requests;

To maintain login state, dev use:

*session + cookie+ JWT token

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RESTFull API: REST stands for Reperesentational State Transfer
resources

/product
/users
/orders

REST CRUD

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RESR principles

Each request is independent

Fontend and Backend are reparated

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JSON comunication

REST APIs usually communicate using JSON

REST Flow:

React -> Axios http request -> Spring rest API -> Json response -> React UI

JSON object

```
{  
    name:"laptop",  
    pruce: 1000,  
}
```

json array

```
[  
    {"id":1},  
    {"id:2},  
]
```

JACKSON

Spring boot uses Jackson to convert: Java Object <-> JSON
using ORM mapping

Class (table); Object(row), Field(Column)

With ORM:

Hibernate automatically generates SQL

JPA = Java Persistence API

Hibernate is the most popular implementation of JPA

Note: Hibernate stores objects in memory and tracks their states;

Hibernate automatically detechs object changes

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Lazy Loading (data is loaded only when needed)

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Spring:

Spring manages objects instead of devs manually creating them.

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Bean = object managed by Spring

?? stored

IOC container??? (inversion of control)

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frontend:

useState:

when state changes, React re-renders the UI;

React render flow:

State change -> Virtual DOM -> Diffing algorithm -> Real DOM update

(note: Virtual DOM is an in-memory copy of the real DOM.

It improves rendering performance)

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React lifecycle:

mount: Component is created for the first time (useEffect)

update: Occurs when state or props change

unmount: component is destroyed

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Async javascript:

-JS is single-threaded

-Async programming prevents UI blocking

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Architect App:

Client -> Dispatcher Servlet -> Controller -> service -> repository -> DB

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Detail:

React State -> Axios http request -> REST API -> Controller -> Service

-> Repository -> Hibernate ORM -> MySQL -> JSON response -> React Re-render

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